

Agent-guided hierarchy: from captured evidence to grounded retrieval

ONE-TIME MAP CONSTRUCTION

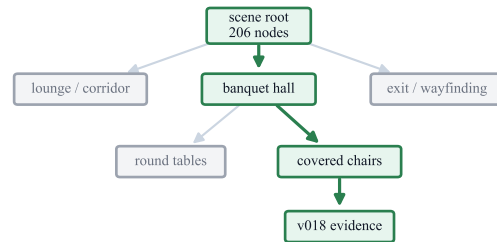
A Posed RGB-D capture



B Semantic ViewJSON record



C Deterministic hierarchy assembly



five scenes: 97 views | 1,066 items | 66,937 input / 179,817 tree token-equiv.

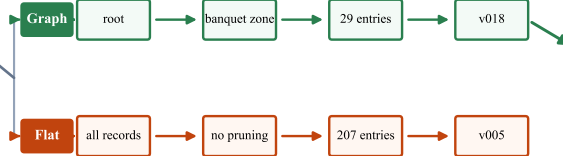
PER-QUERY RETRIEVAL

D Natural-language query



target covered chairs
attribute white

E Graph-pruned vs. flat traversal



views: 3 vs 20 | estimated context: 555 vs 3,110 tokens

F Grounded evidence



graph hit@1: yes | 2.07x speedup | 82.2% fewer est. tokens